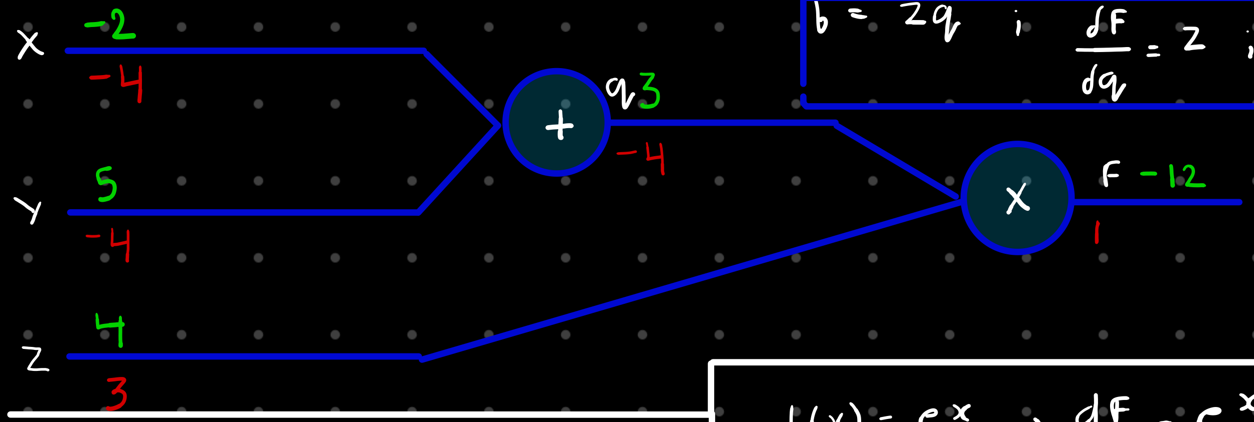
Simple example

$f(x,y,z) = (x+y)z$

$$\boxed{q = x+y \;;\; \frac{\delta q}{\delta x} = 1 \;;\; \frac{\delta q}{\delta y} = 1}$$

$$\boxed{f = zq \;;\; \frac{\delta F}{\delta q} = z \;;\; \frac{\delta F}{\delta z} = q}$$



$f(x) = e^x \longrightarrow \frac{dF}{dx} = e^x$

$f_a(x) = ax \longrightarrow \frac{dF}{dx} = a$

$f(x) = \frac{1}{x} \longrightarrow \frac{dF}{dx} = -\frac{1}{x^2}$

$f_c(x) = c+x \longrightarrow \frac{dF}{dx} = 1$

Another example

$$f(x,w) = \frac{1}{1+e^{-(w_0 x_0 + w_1 x_1 + w_2)}}$$